HeLLO 3AI

SECTION A: BASIC PROGRAM STRUCTURE & I/O (Questions 1-5)

Problem 1: Personal Information Display Write a Java program that declares variables to store your name (String), age (int), and favorite character (char). Print these values in a formatted manner:

Name: [Your Name]
Age: [Your Age]
Favorite Character: [Char]

Problem 2: Command Line Calculator Write a program that accepts two integers and one operator (+, -, *, /) as command line arguments (String[] args). Perform the operation and display the result. Handle division by zero with an appropriate message.

Problem 3: Temperature Converter Write a program to convert Celsius to Fahrenheit using the formula: F = (C × 9/5) + 32. Accept temperature as a command line argument. Use Double.parseDouble() to convert the argument.

Problem 4: Student Marks Analyzer Create variables for marks in 5 subjects (byte data type to save memory). Calculate and display:

- Total marks
- Average (as float)
- Percentage (as double)

Problem 5: ASCII Value Printer Write a program that takes a character input (using command line arguments) and prints its ASCII value. Then print the next 5 consecutive characters and their ASCII values.

SECTION B: OPERATORS & CALCULATIONS (Questions 6-11)

Problem 6: Simple Interest Calculator Calculate simple interest using the formula: SI = (P × R × T) / 100. Accept Principal (long), Rate (float), and Time (int) using appropriate data types to minimize memory usage. Display the result with 2 decimal places.

Problem 7: Bitwise Swap Write a program to swap two integers without using a temporary variable or arithmetic operators. Use only bitwise XOR (^) operator. Display values before and after swapping.

Problem 8: Compound Assignment Demonstrator Initialize an integer variable num = 100. Perform the following operations in sequence using compound assignment operators only (+=, -=, *=, /=, %=):

- Add 50
- Subtract 30
- Multiply by 2
- Divide by 5
- Modulus by 7
- Print the value after each operation.

Problem 9: Circle Measurements Given the radius as a double, calculate and display:

- Area (πr^2) - use 3.14159
- Circumference ($2\pi r$)
- Diameter ($2r$)
- Use the ternary operator to check if radius is positive; if not, print "Invalid radius".

Problem 10: Increment/Decrement Sequence Initialize int a = 5 and int b = 10. Write a program that prints the following sequence using pre/post increment/decrement operators:

- Print a++ (then print current a)
- Print ++b (then print current b)
- Print --a + b-- (then print both values)

Problem 11: Logical Gate Simulator Accept three boolean values (true/false) as command line arguments. Simulate an AND gate, OR gate, and NOT gate using logical operators. Display the truth table results for combinations of these inputs.

3

SECTION C: DECISION MAKING (Questions 12-19)

Problem 12: Grade Calculator Accept a percentage (0-100) as input. Use if-else-if ladder to determine the grade:

- 90-100: A
- 80-89: B
- 70-79: C
- 60-69: D

- Below 60: F
- Validate input (reject negative numbers or >100).

Problem 13: Leap Year Checker Accept a year as input. Determine if it's a leap year using these rules:

- Divisible by 4, but not by 100, unless also divisible by 400
- Use nested if statements and logical operators (&&, ||).

Problem 14: Number to Words (Switch) Accept a single digit (0-9) as input. Use a switch statement (traditional or modern arrow syntax) to print the number in words (e.g., input 5 prints "Five"). Include a default case for invalid inputs.

Problem 15: Menu-Driven Calculator Display a menu:

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Modulus
6. Exit
7. Use a switch-case structure. For choice 1-5, accept two numbers and perform the operation. Loop until user selects 6 (use a do-while loop for the menu).

Problem 16: Days in Month Calculator Accept month number (1-12) and year. Use switch-case to determine the number of days in that month. For February, check if it's a leap year to return 28 or 29 days.

Problem 17: Largest of Three (Ternary) Find the largest of three numbers using only nested ternary operators (no if statements). Print the largest number with a message identifying which variable it was.

Problem 18: Vowel or Consonant Checker Accept a single character. Use if-else to determine if it's:

- Vowel (a, e, i, o, u - both cases)
- Consonant
- Digit

- Special Character Use Character wrapper class methods (isDigit(), isLetter()) where appropriate.

Problem 19: Tax Calculator Calculate income tax based on these slabs:

- 0 - 2,50,000: No tax
- 2,50,001 - 5,00,000: 5%
- 5,00,001 - 10,00,000: 20%
- Above 10,00,000: 30%
- Use if-else if ladder. Add 4% health and education cess on the calculated tax.

SECTION D: LOOPS (Questions 20-27)

Problem 20: Multiplication Table Accept a number and print its multiplication table from 1 to 20 using a for loop. Format the output as:

5 x 1 = 5
5 x 2 = 10
...

Problem 21: Factorial Calculator Calculate the factorial of a given number using:

- Method A: for loop
- Method B: while loop
- Handle negative inputs (factorial not defined) and 0! = 1. Use long data type for the result.

Problem 22: Fibonacci Series Print the first n terms of the Fibonacci sequence (0, 1, 1, 2, 3, 5, 8...) where n is input by the user. Use a for loop. Ensure you handle the first two terms separately.

Problem 23: Prime Number Checker Accept a number and determine if it's prime using a loop that runs from 2 to √n (use Math.sqrt()). Optimize the loop to skip even numbers after checking 2.

Problem 24: Number Pyramid Using nested for loops, print this pattern for n=5:

```
    1
   1 2
  1 2 3
 1 2 3 4
1 2 3 4 5
```

Problem 25: Do-While Input Validation Write a program that repeatedly asks the user to enter a positive even number until they comply. Use a do-while loop to ensure the prompt appears at least once. (Simulate user input using a predefined array for testing).

Problem 26: Enhanced For Loop Sum Initialize an int array with 10 values. Use an enhanced for-loop (for-each) to calculate and display:

- Sum of all elements
- Sum of even numbers only
- Average as a double

Problem 27: Reverse Number Accept an integer and print its reverse using a while loop (e.g., input 12345 → output 54321). Handle negative numbers (preserve the sign).

5

SECTION E: ARRAYS & STRINGS (Questions 28-32)

Problem 28: Array Statistics Create an array of 20 integers initialized with random values (use Math.random()). Write methods to find:

- Maximum value
- Minimum value
- Second largest value
- Sum of all elements using enhanced for-loop

Problem 29: String Case Toggle Accept a String and create a new String where every alternate character is uppercase and lowercase (e.g., "hello" → "HeLlO"). Use charAt() and String concatenation in a loop.

Problem 30: Autoboxing/Unboxing Demonstration Create a method that accepts a primitive int array and returns the sum as an Integer object (autoboxing). In main, unbox this Integer to a primitive int and display it. Also demonstrate converting an int to String using wrapper class.

Problem 31: Linear Search Implement linear search on an int array. Return the index if found, -1 otherwise. Use a for-each loop with a break statement when found.

Problem 32: Jagged Array Sum Create a jagged array (array of arrays with different lengths) representing student marks in different subjects:

- Student 1: 3 subjects
- Student 2: 4 subjects
- Student 3: 2 subjects
- Calculate and display the average marks for each student using nested loops.

6

SECTION F: CONTROL FLOW & JUMP STATEMENTS (Questions 33-35)

Problem 33: Continue Skip Pattern Print numbers from 1 to 50, but skip all multiples of 3 using the continue statement. Count how many numbers were printed and display the count.

Problem 34: Labeled Break Search Create a 5x5 matrix (2D array) with numbers 1-25. Search for a specific number using nested loops. If found, use a labeled break to exit both loops immediately and print the position (row, col).

Problem 35: Return Statement Usage Write a method isPerfectNumber(int n) that returns boolean (use return statement). A perfect number equals the sum of its proper divisors (e.g., 6 = 1+2+3). In main, use this method to print all perfect numbers between 1 and 1000. Demonstrate early return if number <= 0.